

Conductor gaps in a Koblitz curve family

Research record | 2026-10-06 Pacific time | exact arithmetic, bounded scan

Question: Find known-source ordinary curve families whose Frobenius order permits a large prime conductor transition. Scan $E_m: y^2 + xy = x^3 + 1$ over $F_{(2^m)}$, $13 \leq m \leq 131$. The source End order is maximal of discriminant -7 .

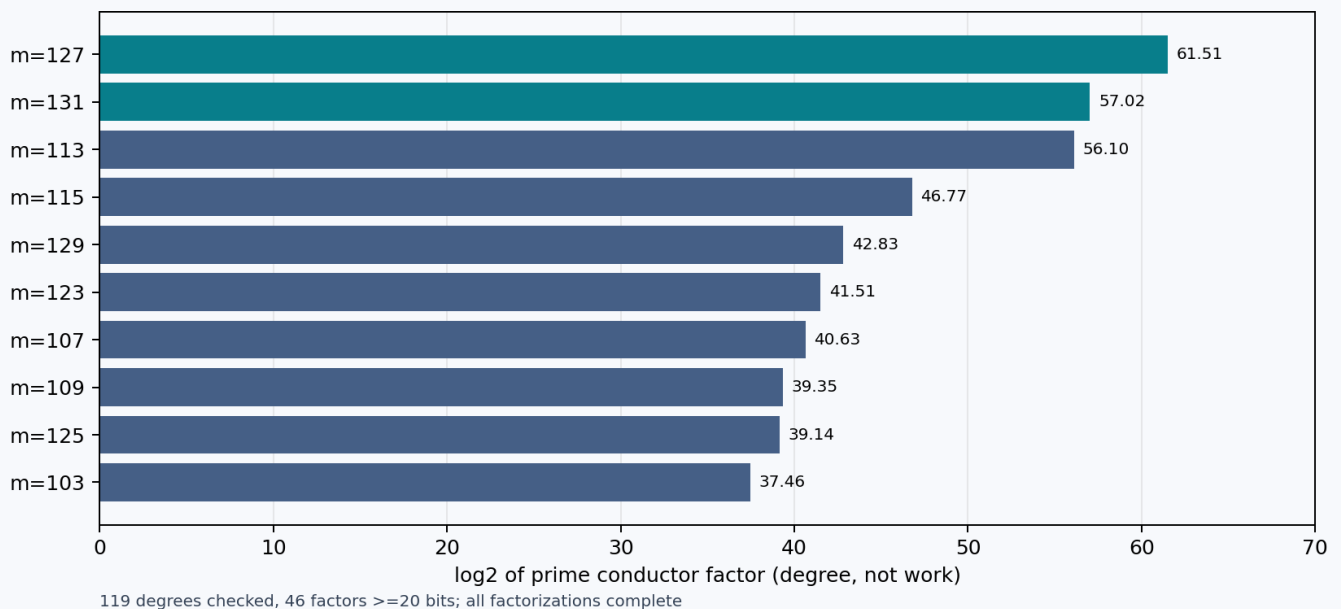
Observed results

119/119 degree rows have complete conductor factorizations verified by recomposition and deterministic prime checks below 2^{64} . 46 factor hits have bit length at least 20. Top hits:

m	largest prime p in f_pi	log2(p)	h(O_p) / h(O_K)
127	3293187233103900007	61.514	3293187233103900006
131	146505763881528721	57.024	146505763881528722
113	77031318395801969	56.096	77031318395801970
115	120145574032529	46.772	120145574032530
129	7798657376119	42.826	7798657376118

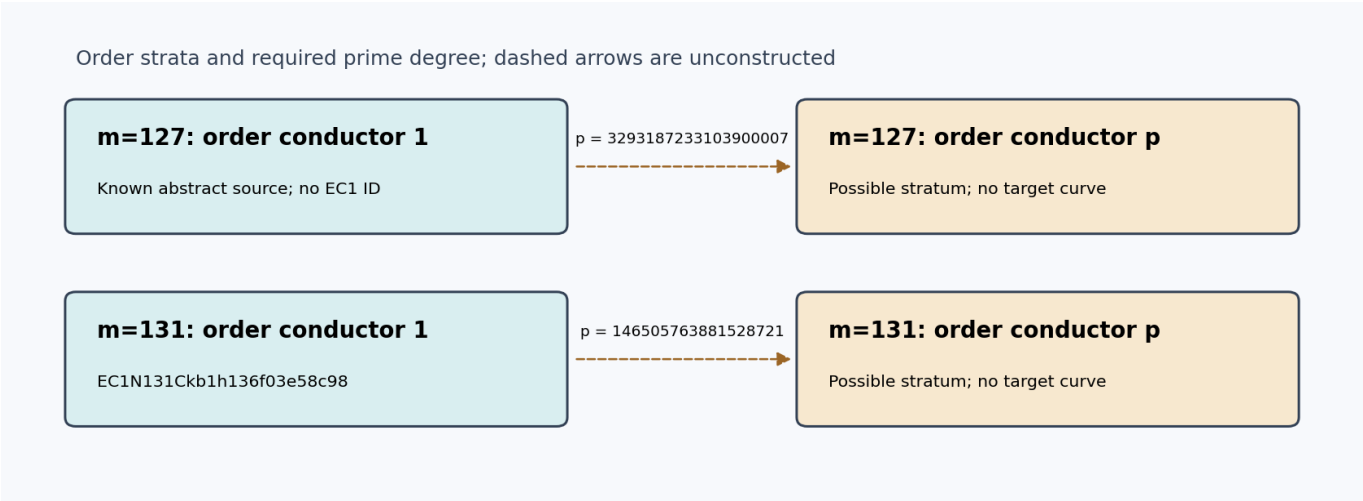
For $m=127$ the prime factor is 3293187233103900007 (about $2^{61.514}$). This is an isogeny degree in the order graph, not a time bound. Its destination order has discriminant $-7p^2$ and class number $p-1$; no destination curve or map was constructed.

Largest conductor factors in the bounded Koblitz scan



What the gap identifies

The source tau satisfies $\tau^2 + \tau + 2 = 0$ over F_2 . The corresponding discriminant -7 is fundamental. For $\tau^m = a_m + b_m \tau$, the exact identity $t_{m^2-4} \cdot 2^m = -7b_m^2$ gives $f_{\pi} = |b_m|$. A prime p dividing f_{π} yields the candidate order $Z + p O_K$; $h(O_p)/h(O_K) = p - (-7/p)$.



Evidence boundaries

The $m=131$ source has exact ID EC1N131Ckb1h136f03e58c98. Other source rows in this scan specify the abstract curve model but no particular field representation or EC1 ID. Destination strata are arithmetic possibilities. Curve models, maps, subgroup preservation, relation cost, and DLP advantage remain unknown.

Scope of completion: all degrees 13..131 in this family were checked. The scan does not survey all elliptic curves or certify that a specific destination across the largest factor has an advantage. The $m=19$ conductor-457 descendants and $m=131$ degree-263 descendants are existing constructed controls.

Source links

Sutherland, Isogeny volcanoes (ordinary levels and orders). Exact $m=131$ curve registry. Prior $m=131$ evidence. Source script and every raw row accompany this PDF.